SIGHT Project - Meeting 15

Project Name: Safety Immersion and Gamified Hazard Training for Industry 5.0 Workers
Project Number: WSIC26-250206-009
Date: February 20, 2026
Time: 11:00 AM EST
Location: Zoom
Meeting Focus: User Diagnostic Survey Results, Research Publication Planning, and Q3 BWC Presentation Preparation

Attendees

- **Miami University (MU):** Fadel Megahed (PI), Arthur Carvalho (Co-PI), Jay Shan (Co-PI), Ibrahim Yousif (Post Doc), Reza Abrishambaf (Senior Personnel), Mohamed Farrag (PM), Michael Wise (GRA), Ryan Singh, Austin Hamilton, Amanda White
- **The Discovery Center:** Maressa Dixon, Kristen Morio, Yue Li
- **University at Buffalo (UB):** Lora Cavuoto (Safety Consultant)
- **MeetKai:** Jacquie Babakanian (Chief of Staff), Andre Amorim, David Arcia, Artur, Declan L, Anushree Banerjee, Vincent Cheong, Alexander Elert, Usuario
- **MaxByte:** Naveen (Technical Lead)

1. Agenda Summary & Approval of Minutes

Administrative Updates: Fadel Megahed opened the meeting by reviewing the minutes from Meeting 14. The team confirmed the approval of the previous minutes and reviewed the status of ongoing action items.

BWC Deliverables & Desk Review: Fadel highlighted that the first **Desk Review** submission is due on **March 04, 2026**. He detailed MU's internal process, which involves a pre-submission review by the university's research office to ensure all documentation meets the BWC's specific formatting and compliance requirements. The team also finalized the main talking points for the Q3 presentation to the BWC on February 26.

2. Industry Partner Updates

A. MeetKai (User Feedback & Asset Development)

- **Diagnostic Survey Integration:** The team discussed the 50+ responses received from the user diagnostic survey. MeetKai is "incubating" this feedback to pivot certain VR elements. Specifically, the data suggested that users found current hazard identification workflows too linear; MeetKai is now looking at more open-world exploration within the modules to simulate real-world movement more accurately.
- **VR/AR Stability:** David Arcia and the engineering team confirmed that the mobile browser compatibility is stable. They highlighted the successful resolution of a complex shader compilation bug in Firefox that was previously affecting visual performance on Android devices, ensuring the simulation runs smoothly across different hardware configurations.
- **3D Asset Pipeline:** Andre Amorim presented updates on the 3D assets, focusing on high-fidelity renders of the pallet trucks and forklift equipment. These assets are being optimized for the "Gamified Engine" to ensure high performance on mobile VR devices. Fadel emphasized the importance of visual clarity on safety labels and instructional signage within the 3D environment to ensure the training remains effective for hazard identification.

B. MaxByte (IIoT Integration & Hardware)

- **Sensor Data Flow:** Naveen provided updates on the IIoT integration. The team is currently working on the communication protocols between the physical sensors on the warehouse floor and the digital twin to ensure low-latency data transmission.
- **Equipment Setup:** Discussion focused on the next steps for the customized pallet truck system (now in the AM Hub) and sensor calibration. Naveen confirmed the hardware for all the needed IIoT systems is at the AM Hub, and the priority is capturing raw sensor data from the three primary machines to build a robust baseline for the AI-driven hazard detection system.

3. Miami University Updates

Technical & Research Updates (Miami University):

- **User Diagnostic Survey:** Lora Cavuoto presented a deep dive into the survey results. Key findings included a significant gap in hazard perception among junior workers compared to seniors, which will serve as the baseline for the "adaptive" part of the gamified training, allowing the difficulty to scale based on user experience.

- **Publication Strategy:**
  - **Arthur and Jay:** Developing a paper for the *Journal of Service Research* (JSR) special issue on "AI-enabled Experience Innovation." The paper will use the SIGHT project as a case study for how AI chatbot integration and real-time feedback improve worker safety training "stickiness" and engagement.
  - **Ibrahim and Amanda:** Revising the manuscript on "Digital Twins for Safety 4.0," detailing the architecture used to sync physical machine states with the VR training environment, specifically focusing on latency requirements for real-time safety alerts and the integration of the pallet truck sensor data.

BWC Q3 Presentation: Fadel reviewed the slide deck for the upcoming BWC progress meeting. The presentation will emphasize the successful integration of the AI chatbot and the Computer Vision (YOLO) module for real-time hazard detection.

4. Action Items

Task	Responsible	Deadline
Present Q3 progress to the BWC	Fadel / MU Team	Feb 26, 2026
Finalize and submit Desk Review 1	MU Team	Mar 04, 2026
Complete Gamified Engine development	MeetKai	Feb 28, 2026
Integrate the analytics framework with VR modules	MeetKai	March 20, 2026
Finalize data acquisition from the three machines	MaxByte	Feb 27, 2026
Confirm IIoT sensor calibration for the pallet truck	MaxByte	Feb 27, 2026
Share raw sensor logs with MU for AI training	MaxByte	Mar 06, 2026
Finalize 3D equipment renders	MeetKai / MU	March 15, 2026

5. Wrap-Up & Next Meeting

Next Meeting: Bi-monthly whole team meeting to be scheduled for March 06, 2026.